The “ARIES Genomics” Italian public health surveillance system

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Introduction

The scope of the project was to develop an Information System for the collection of genomic and epidemiological data to enable the Next Generation Sequencing (NGS)-based surveillance of infectious epidemics, foodborne outbreaks and diseases at the animal-human interface. The system was expected to offer the ability to be used with a low learning curve directly by the persons target of the results of the analyses, keeping the chain of information exchange as short as possible avoiding data transmission delays. Therefore, the user interface had to be simple yet complete, producing essential comprehensive outcomes, together with the complete data available for users with bioinformatic skills.

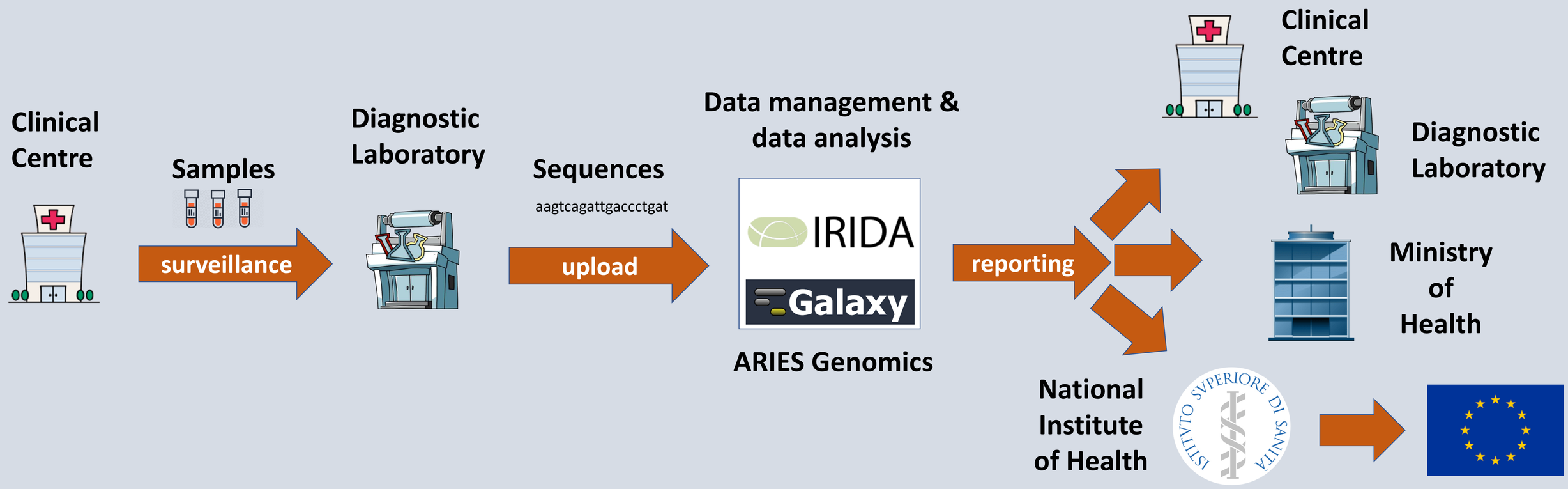
The system was built engaging two personalised open source platforms: A **Galaxy** instance [1] implemented as a cluster (Advanced Research Infrastructure for Experimentation in genomicS, **ARIES**, [2]) and an **IRIDA** (Integrated Rapid Infectious Disease Analysis, [3]) instance.

IRIDA provides the data management and the possibility to launch analysis pipelines that are executed on the ARIES cluster.

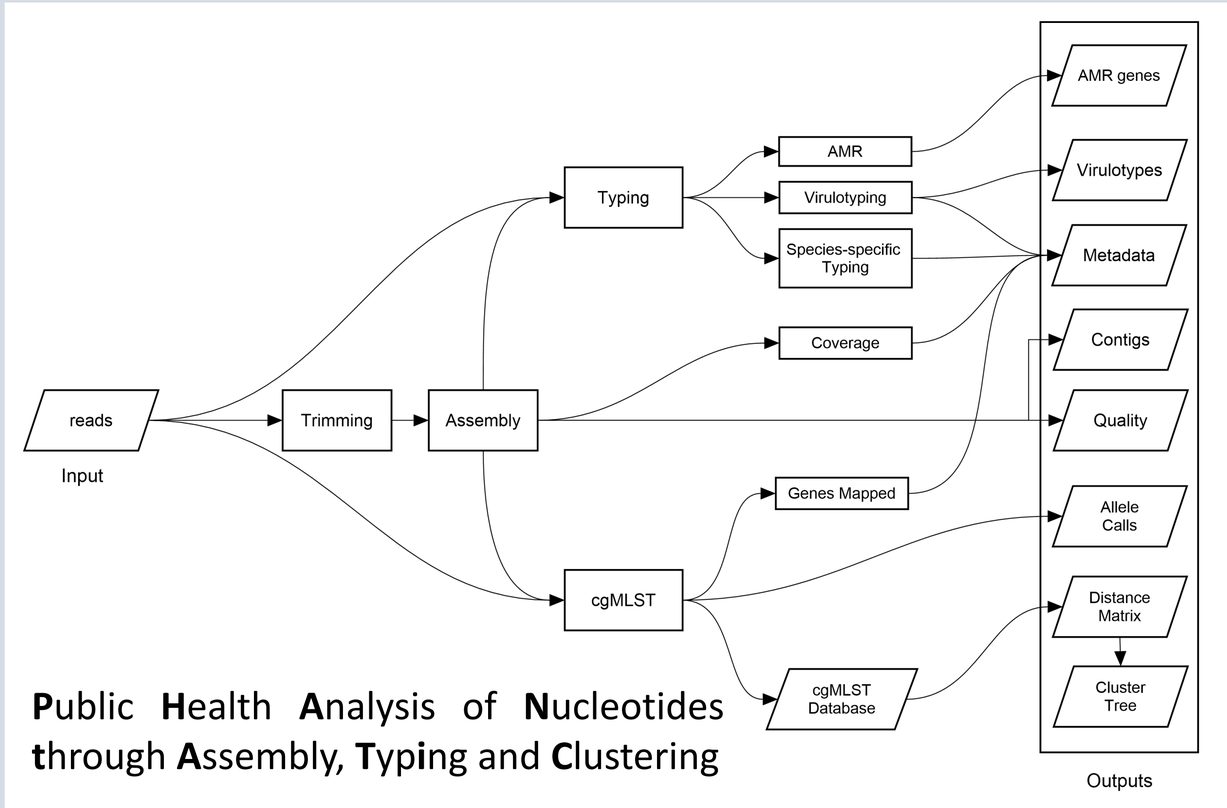
Personalisations

- Translation in Italian of the IRIDA user interface.
- Possibility in IRIDA to start analyses with a combination of paired-end and single-end reads. Possibility in ARIES to accept them.
- Samples are sent to ARIES as a list instead of as a series of single data items.
- Upon creation of a sample, metadata is retrieved from external sources.
- The pipeline that is automatically launched upon upload of sample sequences was defined to be PHANtAsTiC performing typing and clustering elaborations.
- At the conclusion of the automated pipeline, an e-mail is sent to Project members containing concise information on the clustering results (whether the sample is part of a cluster) and in case of a cluster also to the other Regions involved in the cluster, the Ministry of Health and the Istituto Superiore di Sanità (National Institute of Health).
- Four analysis pipelines were added to the platform:
  1. cgMLST analysis (MentaLiST); 2. SNP analysis (PopPUNK); 3. Creation of a summary report; 4. Multi virulotyping (calculation of a matrix of samples-virulence genes).
- Addition of a Participant system role that can view results but cannot export data.
- Definition of a dedicated master National Project for every surveillance acting as a general container for data from the Regional Projects.
- Sensitive data from the Regional Projects is not shown in the National Project.

Information Flow



PHANtAsTiC pipeline



Public **H**ealth **A**nalysis of **N**ucleotides **t**hrough **A**ssembly, **T**yping and **C**lustering

Conclusions

The integrated platform is currently hosting the Italian national surveillance system for infections by *Listeria monocytogenes* and the local surveillance system for infections by Shigatoxin-producing *Escherichia coli* (STEC) and counts 70 users, mainly belonging to the Regional Public Health Services.

The system has been well-accepted by all different types of users because it has proven intuitive enough for those without particular computer skills, while yet powerful for the needs of the users with advanced bioinformatic experience.

The system has made the transition from PFGE-based to NGS-based surveillance fluid, allowing concomitantly a better overview of the existence of clusters with respect to geographical location as well as to temporal occurence.

References

[1] https://galaxyproject.org
[2] https://doi.org/10.1101/2020.05.14.095901
[3] https://irida.ca